

Assessment: Flutter UI Design Challenge – “Serenity” Wellness Dashboard

Objective

Design and implement **only the user interface** for the **Home Screen** and a **Habit Detail Bottom Sheet** of a fictional wellness app called *Serenity*. No backend logic, no real data handling—hardcoded sample data is allowed. The goal is to demonstrate mastery of Flutter’s layout, custom painting, animation, and interaction design capabilities.

Your implementation must follow the exact specifications below. The complexity is intentionally high to test advanced UI skills; superficial AI-generated code will not meet all the detailed requirements.

Scenario

Serenity is a daily wellness companion. The Home Screen gives users an at-a-glance view of their emotional state, daily goal progress, motivational content, and an interactive list of wellness habits they can explore. Tapping a habit opens a partially overlapping, blur-backed bottom sheet with detailed statistics and a quick-log action. The entire experience is fluid, with custom-painted graphics, 3D effects, and coordinated spring animations.

Design Specifications

Colour Palette

Role

Hex Code

Primary Background

 #F9F7F7 (soft warm white)

Role	Hex Code
Surface Cards	#FFFFFF with 8% opacity black shadow
Primary Accent	#6C63FF (vibrant purple)
Secondary Accent	#FF6584 (coral pink)
Success	#2EC4B6
Warning	#FFBF46
Text Primary	#2D2D2D
Text Secondary	#7A7A7A
Glassmorphism (bottom sheet)	rgba(255,255,255,0.85) with BackdropFilter blur 12 px

Typography

- **App title / large numbers:** Google Fonts `Playfair Display`, weight 700, letterSpacing - 0.5
- **Headings (card titles, section headers):** `SF Pro Rounded` (or `Cupertino` equivalent), weight 600
- **Body, labels, captions:** `Inter`, weight 400
- **Mood ring text:** `Dancing Script`, weight 700, italic

Spacing & Dimensions

- Screen horizontal padding: 24 dp
- Card corner radius: 20 dp (outer), 14 dp (inner elements)
- Circular progress indicator: diameter 160 dp, stroke width 14 dp
- Habit grid cross-axis count: 2, childAspectRatio: 0.85
- Habit card 3D tilt max rotation: $\pm 12^\circ$ (x) and $\pm 8^\circ$ (y), triggered by scroll position relative to screen center
- Bottom navigation bar height: 72 dp, with a top-notch cutout of radius 36 dp for a floating action button
- Bottom sheet default peek height: 180 dp, fully expanded: 65% of screen height

- Avatar size in bottom sheet: 70 dp, overlapping the sheet's top edge by 35 dp
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Screen Layout & Required Elements

1. App Bar

- Transparent background, no shadow.
- Left: circular avatar (40 dp, image placeholder `assets/avatar.png`).
- Right: a **bell icon** with a subtle red dot badge (6 dp radius) when "unread" (hardcoded as true).
- Center: current date in `Inter`, size 12, Secondary Text colour, format "Wednesday, 3 Jun".

2. Hero Section

A vertically scrollable column that includes:

a) Mood Ring (Custom Painter)

- A circle of diameter 120 dp.
- The circle's stroke is a **conic gradient** that rotates smoothly as the user drags a thumb along the ring. The gradient colours transition from `#FF6584` (stress) → `#FFBF46` (neutral) → `#2EC4B6` (calm) → `#6C63FF` (focused).
- A glowing shadow (blur radius 18, colour matches current gradient midpoint with 40% opacity) must radiate outward.
- Inside the ring, display the **current mood word** in `Dancing Script` (size 18, Primary Accent).
Implement this using `GestureDetector` to convert drag angle into a 0–1 value, but no real data storage required; just update a local state variable.
- Small label below: "How are you feeling?" in `Inter` 12, Secondary Text.

b) Daily Goal Progress

- A custom-painted **circular progress indicator** (diameter 160 dp) with the following features:
 - Background track: `#E0E0E0`, width 14 dp, rounded caps.
 - Progress arc: **sweep gradient** from Primary Accent to Secondary Accent, width 14 dp, rounded caps.

- A **glow effect** on the progress arc (shadow with same gradient, blur 8, spread 2).
- Center text: "78%" in Playfair Display 32, Primary Text, and below "of daily goal" in Inter 11, Secondary Text.
- The progress value is hardcoded at 0.78, but the arc must be drawn with a **smooth** `AnimationController` that animates from 0 to 0.78 on first build (duration 1.2 s, `Curves.easeOutCubic`).

3. Motivational Quote Carousel

- A horizontal `PageView` with three quote cards.
- Each card is 85% of screen width, height 100 dp, rounded 20 dp, Surface colour, with a subtle gradient overlay (`LinearGradient` from top-left Primary Accent 10% to transparent).
- Content: quote in Playfair Display italic 16, author in Inter 12 Secondary.
- While scrolling, the current card scales to 1.0, the adjacent cards scale to 0.92 and have a slight vertical offset (+6 dp) creating a **parallax stack effect**.
- Auto-advance every 4 seconds, looping infinitely (use `Timer` but no state logic beyond UI).

4. Habit Categories

- A horizontal scrolling list of **7 circular category chips** (diameter 60 dp).
- Each chip has an icon (use `Icons` or custom assets), a gradient background (Primary to Secondary), and a label below (Inter 10, Secondary Text).
- The selected chip has a **scale animation** (1.0 → 1.15) and a white border of 2 dp. Only one chip can be selected at a time (hardcode default to index 2).
- Selection change triggers the habit grid (see below) to cross-fade to a new set of cards (use `AnimatedSwitcher` with `FadeTransition`).

5. Habit Grid with 3D Tilt

- A 2-column grid showing habit cards (hardcode 4 habits: "Meditation", "Hydration", "Reading", "Stretching").
- Each card (width: fill, height determined by `childAspectRatio` 0.85) has:
 - A **3D perspective tilt** based on its vertical position in the scroll viewport. As the card approaches the center of the screen, it rotates towards the user: X rotation from 0° (top/bottom) to -12° (center), Y rotation from -8° (left) to +8° (right). Use `Transform` with a `Matrix4` and a `ScrollController` listener.
 - Background: Surface colour, rounded 20 dp, with an inner shadow at the bottom.
 - Gradient overlay at top for the habit icon (32 dp) and title (Inter 18, Primary Text).
 - A **daily streak fire emoji** and number (e.g., "🔥 12") at bottom right.

- On tap, the card triggers a **hero animation** (tag: `habit-{name}`) that flies to the bottom sheet's avatar position (coordinate match required). The bottom sheet opens simultaneously.

6. Bottom Navigation Bar

- A custom curved container with a **top notch** cutout (radius 36 dp) at the center for a **Floating Action Button**.
- The FAB is 56 dp diameter, Primary Accent, with a "+" icon, and a drop shadow (blur 12, offset 0,4, colour black 25%).
- Three navigation items on each side: `home`, `insights`, `profile` icons, evenly spaced, with the active icon tinted Primary Accent and scaled 1.2.
- The entire bar has a frosted glass effect: `BackdropFilter blur 8, background rgba(255,255,255,0.7)`.

7. Habit Detail Bottom Sheet

- Triggered when a habit card is tapped. The sheet **peeks** at 180 dp and can be dragged to 65% of screen height.
 - The sheet has a **hero avatar** (the habit icon inside a 70 dp circle) that starts at the tapped card and animates to a position centred horizontally, overlapping the sheet's top edge by 35 dp.
 - Sheet surface: frosted glass (`BackdropFilter blur 12, rgba(255,255,255,0.85)`), rounded top corners 30 dp.
 - Content (scrollable):
 - Habit name in Playfair Display 24.
 - "Current Streak" in Inter 14 Secondary, with a flame icon and large number (Inter 36, Success).
 - A **custom painted line chart** showing the last 7 days (hardcoded values). The line must be a gradient stroke (Primary to Secondary), 3 dp width, with circular data points (radius 4, white fill, coloured border). Grid lines in `#E0E0E0` dashes. The chart must animate from left to right on appear (1 s, `Interval` curve).
 - A "Quick Log" button: pill shaped, full width, gradient from Primary to Secondary, text "Mark Today" in Inter 16 white. On tap, a **confetti burst animation** (use a simple custom painter with 20 small circles of random colours that fly outward and fade) plays for 800 ms – purely visual, no state change needed.
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Animations & Micro-interactions (Mandatory)

1. **Mood ring glow** pulsates subtly (opacity oscillation between 0.3 and 0.6 over 2 seconds, looping).
 2. **Circular progress** arc fills with easeOutCubic on first load.
 3. **Quote carousel** auto-advances with a 400 ms easeInOut slide and scaling of adjacent cards.
 4. **Category chip** selection scales with a spring effect (SpringSimulation, stiffness 200, damping 15).
 5. **Habit card 3D tilt** updates at 60 fps, but clamped and smoothed (use `lerp` with 0.15 factor to avoid jitter).
 6. **Hero animation** from habit card to bottom sheet avatar (standard Flutter hero, ensure unique tags).
 7. **Line chart** draws path progressively.
 8. **Confetti burst** on button tap.
 9. Pull-to-refresh on the entire home screen: when pulled down > 80 dp, the standard refresh indicator is **replaced** by a custom “morphing” loader – a circle that splits into three smaller circles, rotates, and reforms. Use `CustomPaint` and an `AnimationController`. After 2 seconds, it collapses back.
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Custom Widgets to Build (Reusable)

- `MoodRing`
 - `GlowCircularProgress`
 - `HabitCard3D` (accepts title, icon, streak, heroTag)
 - `MorphingRefreshIndicator` (wrapper that modifies `RefreshIndicator` behaviour)
 - `CurvedBottomNavBar`
 - `AnimatedLineChart`
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Assets (Placeholders)

You must include placeholder assets in your project:

- `assets/avatar.png`
- `assets/quote_bg_1.png` (optional, can use a coloured container)

- All icons can be from `Icons` class, but the final UI must closely match the described appearance.
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Constraints

- **State management:** Use `setState` or a simple `ChangeNotifier` – no external packages for state.
 - **No backend, no API calls.** All data is hardcoded in a mock data file.
 - **Flutter version:** Stable 3.22 or newer.
 - **Only UI implementation is evaluated;** pressing buttons may trigger animations but must not navigate to new screens (except the bottom sheet).
 - **AI code generation tools are not prohibited,** but the assignment is designed to be extremely difficult to generate fully with a simple prompt because of the highly specific, interconnected custom painters, animations, and micro-interactions.
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Submission

1. Complete Flutter project (zipped) with all Dart files, assets, and `pubspec.yaml`.
 2. A short screen recording (GIF or MP4) demonstrating:
 - Scrolling the home screen, 3D tilt effect visible.
 - Mood ring drag interaction and glow pulsation.
 - Category chip selection and grid cross-fade.
 - Hero animation to bottom sheet.
 - Pull-to-refresh morphing loader.
 - Confetti burst.
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Evaluation Rubric (Total: 100 points)

Category	Points	Details
Layout Fidelity	25	Exact match to spacing, sizing, and composition specifications.
Custom Painting	25	Quality of MoodRing, GlowCircularProgress, AnimatedLineChart, MorphingRefreshIndicator.
Animations & Micro-interactions	25	Smoothness, correct curves, 3D tilt math, hero transitions, pulsing glow, confetti, etc.
Code Structure & Reusability	15	Clean widget separation, custom widget reusability, no excessive rebuilds.
Overall Polish & Creativity	10	Attention to detail (shadows, typography, glassmorphism), creative flair within constraints.